

Newton's polynomial

Newton's interpolation polynomial is a method for constructing a polynomial that passes through a given set of data points. It is built incrementally using divided differences.

Steps to Construct Newton's Interpolation Polynomial

- Given Data Points:

Suppose that we have data points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$.

- Compute Divided Differences:

- Zero-order divided difference:

$$f[x_i] = y_i$$

- First-order divided difference

$$f[x_i, x_{i+1}] = \frac{f[x_{i+1}] - f[x_i]}{x_{i+1} - x_i}$$

- Second-order divided difference:

$$f[x_i, x_{i+1}, x_{i+2}] = \frac{f[x_{i+1}, x_{i+2}] - f[x_i, x_{i+1}]}{x_{i+2} - x_i}$$

- Continue this process up to n -th order.

- Construct the Polynomial:

The Newton's interpolation polynomial is:

$$\begin{aligned} P(x) = & f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + \dots \\ & \dots + f[x_0, x_1, \dots, x_n](x - x_0)(x - x_1) \cdots (x - x_{n-1}) \end{aligned}$$

Problem

Consider the table:

x	2	2.5	4
$f(x)$	0.5	0.4	0.25

To find the Newton's interpolation polynomial $P(x)$:

1. Compute the Divided Differences:

$$\begin{aligned} f[2] &= 0.5 \\ f[2, 2.5] &= 0.4 \\ f[4] &= 0.25 \end{aligned}$$

First-order divided differences:

$$\begin{aligned} f[x_i, x_{i+1}] &= \frac{f[x_{i+1}] - f[x_i]}{x_{i+1} - x_i} \\ f[2, 2.5] &= \frac{0.4 - 0.5}{2.5 - 2} = -0.2 \\ f[2.5, 4] &= \frac{0.25 - 0.4}{4 - 2.5} = -0.1 \end{aligned}$$

Second-order divided difference:

$$\begin{aligned} f[x_i, x_{i+1}, x_{i+2}] &= \frac{f[x_{i+1}, x_{i+2}] - f[x_i, x_{i+1}]}{x_{i+2} - x_i} \\ f[2, 2.5, 4] &= \frac{-0.1 - (-0.2)}{4 - 2} = 0.05 \end{aligned}$$

In short:

x	$f[x_i]$	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$
2	0.5		
2.5	0.4	$\frac{0.4-0.5}{2.5-2} = -0.2$	
4	0.25	$\frac{0.25-0.4}{4-2.5} = -0.1$	$\frac{-0.1-(-0.2)}{4-2} = 0.05$

2. Construct the Polynomial:

$$\begin{aligned} P(x) &= f[2] + f[2, 2.5](x - 2) + f[2, 2.5, 4](x - 2)(x - 2.5) \\ &= 0.5 - 0.2(x - 2) + 0.05(x - 2)(x - 2.5) \end{aligned}$$

3. Simplify the Polynomial:

$$\begin{aligned}
 P(x) &= 0.5 - 0.2(x - 2) + 0.05(x - 2)(x - 2.5) \\
 &= 0.5 - 0.2x + 0.4 + 0.05(x^2 - 4.5x + 5) \\
 &= 0.5 - 0.2x + 0.4 + 0.05x^2 - 0.225x + 0.25 \\
 &= 0.05x^2 - 0.425x + 1.15
 \end{aligned}$$

Therefore, the Newton's interpolation polynomial is:

$$P(x) = 0.05x^2 - 0.425x + 1.15$$